Introduction to the Finite Element Method

Lecture notes

Time Dependent 1D problems - Problem introduction

In the following, we use the FEM methods for spatial discretization and the finite difference method for time discretization.
Consider the Balance of Linear momentum equation in 1D:

$$
\begin{aligned}
\text{\grad} \cdot \sigma + f(x,t) &= \rho_0 \frac{\partial^2 u}{\partial t^2} \\
&= \rho_0 \frac{\partial v}{\partial t} \\
&= \rho_0 a(x,t)
\end{aligned}
$$

where: - $\sigma(x,t)$ is the stress

- $f(x,t)$ is the body force
- ρ_0 is the density
- u is the displacement
- v is the velocity
- a is the acceleration

Time Dependent 1D problems - single point in space

For a specific point in space, we can write the equation as:

$m\dot{v} = F = ma \quad \text{(I)}$

where all the forces acting on the point are lumped into a single force F and the mass is $m = \rho_0 \Delta x$.

In the following, we will use the Taylor expansion around the time $t + \theta\Delta t$ to expand the quantities of interest at times t and $t + \Delta t$ where θ is a parameter that can be chosen to optimize the time integration scheme.

- For the velocities:

$$v(t+\Delta t) = v(t+\theta\Delta t) + \frac{dv}{dt}\Big|_{t+\theta\Delta t}(1-\theta)\Delta t + \frac{1}{2}\frac{d^2v}{dt^2}\Big|_{t+\theta\Delta t}(1-\theta)^2(\Delta t)^2 + O(\Delta t)^3 \quad (2.1)$$

$$v(t) = v(t+\theta\Delta t) - \frac{dv}{dt}\Big|_{t+\theta\Delta t}\theta\Delta t + \frac{1}{2}\frac{d^2v}{dt^2}\Big|_{t+\theta\Delta t}\theta^2(\Delta t)^2 + O(\Delta t)^3 \quad (2.2)$$

- For the position:

$$u(t+\Delta t) = u(t+\theta\Delta t) + \frac{du}{dt}\Big|_{t+\theta\Delta t}(1-\theta)\Delta t + \frac{1}{2}\frac{d^2u}{dt^2}\Big|_{t+\theta\Delta t}(1-\theta)^2(\Delta t)^2 + O(\Delta t)^3 \quad (2.3)$$

$$u(t) = u(t+\theta\Delta t) - \frac{du}{dt}\Big|_{t+\theta\Delta t}\theta\Delta t + \frac{1}{2}\frac{d^2u}{dt^2}\Big|_{t+\theta\Delta t}\theta^2(\Delta t)^2 + O(\Delta t)^3 \quad (2.4)$$

- For the Force we obtain a weighted average of the forces at the two time steps:
  $$F(t+\theta\Delta t) = \theta F(t+\Delta t) + (1-\theta)F(t) \quad (2.5)$$
- Substracting Equations (1) and (2) we obtain:
  $$\frac{dv}{dt}\Big|_{t+\theta\Delta t} = \frac{v(t+\Delta t) - v(t)}{\Delta t} + \hat{O}(\Delta t) \quad (2.6)$$
  where $\hat{O}(\Delta t)$ is the error term that depends on the choice of θ, for $\theta = 1/2$ we obtain $\hat{O} = O(\Delta t)^2$ and otherwise we get $\hat{O} = O(\Delta t)$.
  By using equation 6 and inserting it into equation (I) we obtain:
  $$v(t+\Delta t) = v(t) + \frac{\Delta t}{m}F(t+\theta\Delta t) + \hat{O}(\Delta t)^2 \quad (2.7)$$
- If we take the weighted sum of (2.1) and (2.2) we obtain:
  $$v(t+\theta\Delta t) = \theta v(t+\Delta t) + (1-\theta)v(t) + O(\Delta t)^2 \quad (2.8)$$
- We can do the same for the positions and obtain:
  $$\frac{u(t+\Delta t) - u(t)}{\Delta t} = v(t+\theta\Delta t) + \hat{O}(\Delta t) \quad (2.9)$$
  Which combined with equation (2.8) gives us:
  $$u(t+\Delta t) = u(t) + (\theta v(t+\Delta t) + (1-\theta)v(t))\Delta t + \hat{O}(\Delta t)^2 \quad (2.10)$$
  Now, we can use equations (2.10) and (2.7) to obtain
  $$u(t+\Delta t) = u(t) + v(t)\Delta t + \frac{\theta(\Delta t)^2}{m}F(t+\theta\Delta t) + \hat{O}(\Delta t)^2 \quad (2.11)$$

- When $\theta = 1$ we obtain an implicit integration scheme (backward Euler) and when $\theta = 0$ we obtain an explicit integration scheme (forward Euler).
- The backward Euler scheme is unconditionally stable, while the forward Euler scheme is conditionally stable and requires a small time step to ensure stability.
- For both we obtain $\hat{O(\Delta t)^2} = O(\Delta t)^2$.
- For $\theta = 1/2$ we obtain the “midpoint”” rule which is unconditionally stable and $\hat{O(\Delta t)^2} = O(\Delta t)^3$.

Time Dependent 1D problems - FEM Formulation

The continuum Equation we try to solve is:

$$\rho_0 \frac{\partial^2 u}{\partial t^2} = \rho_0 \frac{\partial v}{\partial t} = \nabla \cdot \sigma + f(x,t) \quad (3.0)$$

We will make our life easier by defining $F = \nabla \cdot \sigma + f(x,t)$

Using the equations we derived above we can reformulate the problem as

$$\rho_0 v(t+\Delta t) = \rho_0 v(t) + \Delta t(\theta F(t+\Delta t) + (1-\theta)F(t)) \quad (3.1)$$

Multiplying by the test function and integrating by parts we obtain:

$$\int_\Omega v \cdot \rho_0 v(t+\Delta t)\, d\Omega = \int_\Omega v \cdot \rho_0 v(t)\, d\Omega + \Delta t \int_\Omega v \cdot (\theta \Psi(t+\Delta t) + (1-\theta)\Psi(t))\, d\Omega \quad (3.2)$$

And after applying the divergence theorem and enforing $v = 0$ on the boundary we obtain:

$$\begin{aligned}\int_\Omega v \cdot \rho_0 v^{t+\Delta t}\, d\Omega = &\int_\Omega v \cdot \rho_0 v^t\, d\Omega \\ &+ \Delta t\theta\left(-\int_\Omega \nabla v : \sigma\, d\Omega + \int_{\Gamma_t} v \cdot (\sigma \cdot n)\, dA + \int_\Omega v \cdot f\, d\Omega\right)^{t+\Delta t} \\ &+ \Delta t(1-\theta)\left(-\int_\Omega \nabla v : \sigma\, d\Omega + \int_{\Gamma_t} v \cdot t^*\, dA + \int_\Omega v \cdot f\, d\Omega\right)^{t} \quad (3.3)\end{aligned}$$

Plugging in our shape function, and defining the standard matrices with $\{a\}$ being the coeeficients vector, we obtain:

$$[M]\{a\}^{t+\Delta t} = [M]\{a\}^{t} + (\Delta t\theta)\left(-[K]\{a\}^{t+\Delta t} + \{R_f\}^{t+\Delta t} + \{R_t\}^{t+\Delta t}\right)$$
$$+ \Delta t(1-\theta)\left(-[K]\{a\}^{t} + \{R_f\}^{t} + \{R_t\}^{t}\right) \qquad (3.4)$$

where we use

$$\{a\}^{t+\Delta t} = \{a\}^{t} + \Delta t(\theta\{\dot{a}\})^{t+\Delta t} + (1-\theta)\{\dot{a}\}^{t} \qquad (3.5)$$

- For $\theta = 1$ we get the implicit backward Euler scheme:

$$\left([M]\{\ddot{a}\}^{t+\Delta t} + \Delta t[K]\{a\}^{t+\Delta t}\right) = [M]\{\ddot{a}\}^{t} + \Delta t\left(\{R_t\}^{t+\Delta t} + \{R_f\}^{t+\Delta t}\right) \qquad (3.6)$$

where we use

$$\{a\}^{t+\Delta t} = \{a\}^{t} + \Delta t(\{\dot{a}\})^{t+\Delta t} \qquad (3.7)$$

- For $\theta = 0$ we get the explicit forward Euler scheme:

$$\{\dot{a}\}^{t+\Delta t} = \{\dot{a}\}^{t} + \Delta t[M]^{-1}\left(-[K]\{a\}^{t} + \{R_f\}^{t} + \{R_t\}^{t}\right) \qquad (3.8)$$

where we use

$$\{a\}^{t+\Delta t} = \{a\}^{t} + \Delta t\{\dot{a}\}^{t} \qquad (3.9)$$

Some Mathematical preliminaries

Looking at the following equation:

$div(\mathbf{K} \cdot grad(T))$

if $\mathbf{K}$ is a constant multiplied bu the identity matrix, we can write it as:

$$div(\mathbf{K} \cdot grad(T)) = div(\mathbf{K}\left[\frac{\partial T}{\partial x}\mathbf{e}_1 + \frac{\partial T}{\partial y}\mathbf{e}_2 + \frac{\partial T}{\partial z}\mathbf{e}_3\right]) = \mathbf{K}\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right) = \mathbf{K}\nabla^2 T$$

- Temperature as a scalar field:

$$T \text{ - Scalar} \Rightarrow \text{grad}(T) = \left(\frac{\partial T}{\partial x}\mathbf{e_1} + \frac{\partial T}{\partial y}\mathbf{e_2} + \frac{\partial T}{\partial z}\mathbf{e_3}\right) \text{ - vector}$$

- Conductivity matrix operation:

$$\mathbf{K} \cdot \text{grad}(T) = \begin{bmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{bmatrix} \text{transpose}\left(\left(\frac{\partial T}{\partial x}\mathbf{e_1} + \frac{\partial T}{\partial y}\mathbf{e_2} + \frac{\partial T}{\partial z}\mathbf{e_3}\right)\right)$$

- Defining the heat flux vector:

Let's define $\mathbf{q} = -\mathbf{K} \cdot \text{grad}(T)$ as the heat flux vector.

Divergence of heat flux:

$$\text{div}(\mathbf{q}) = \frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} + \frac{\partial q_z}{\partial z}$$

What happens when we multiply the heat conduction equation by a test function V?

$\text{div}(\mathbf{K}\ \text{grad}(T)) \cdot V =$

This question leads us to the development of the weak form, which is essential for finite element methods.

To answer this, we need to use the **product rule for divergence**:

$\text{div}(V\mathbf{K}\ \text{grad}(T)) = \text{grad}(V) \cdot \mathbf{K} \cdot \text{grad}(T) + V\ \text{div}(\mathbf{K} \cdot \text{grad}(T))$

However, if $\mathbf{K} = k$ is constant, we can simplify this to:

$\text{div}(V\mathbf{K}\ \text{grad}(T)) = k\text{div}(V\ \text{grad}(T)) = k(\text{grad}(V) \cdot \text{grad}(T) + V\nabla^2(T))$

Detailed Expansion

Let's expand $\text{div}(V\mathbf{K}\ \text{grad}(T))$ step by step:

$$\text{div}(V\mathbf{K}\ \text{grad}(T)) = \frac{\partial}{\partial x}\left(VK\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(VK\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(VK\frac{\partial T}{\partial z}\right)$$

Applying the product rule to each term:

$$= \frac{\partial V}{\partial x}K\frac{\partial T}{\partial x} + \frac{\partial V}{\partial y}K\frac{\partial T}{\partial y} + \frac{\partial V}{\partial z}K\frac{\partial T}{\partial z} + VK\frac{\partial^2 T}{\partial x^2} + VK\frac{\partial^2 T}{\partial y^2} + VK\frac{\partial^2 T}{\partial z^2}$$

- We can group these terms into two parts:
- Gradient Terms:

$$\text{grad}(V) \cdot \mathbf{K} \cdot \text{grad}(T) = \left(\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z}\right) \cdot \mathbf{K} \cdot \left(\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z}\right)$$

- For constant K, this becomes:

$$= K\left(\frac{\partial V}{\partial x}\frac{\partial T}{\partial x} + \frac{\partial V}{\partial y}\frac{\partial T}{\partial y} + \frac{\partial V}{\partial z}\frac{\partial T}{\partial z}\right) = A$$

- Second Derivative Terms

  $$V \operatorname{div}(\mathbf{K} \cdot \operatorname{grad}(T)) = VK\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right) = B$$

  The Key Relationship

  From the product rule, we have:

  $$\operatorname{div}(V\mathbf{K} \operatorname{grad}(T)) = A + B$$

  Therefore:

  $$B = \operatorname{div}(V\mathbf{K} \operatorname{grad}(T)) - A$$

  Which gives us:

  $$\operatorname{div}(\mathbf{K} \operatorname{grad}(T))V = \operatorname{div}(V\mathbf{K} \operatorname{grad}(T)) - \operatorname{grad}(V) \cdot \mathbf{K} \cdot \operatorname{grad}(T)$$

  Physical Interpretation

  This mathematical identity is the foundation for:

1. **Integration by parts** in the weak formulation
2. **Converting strong form to weak form** in finite element methods
3. **Moving derivatives from the solution to the test function**

   The term $\operatorname{div}(V\mathbf{K} \operatorname{grad}(T))$ will become a boundary integral when integrated over the domain, while $\operatorname{grad}(V) \cdot \mathbf{K} \cdot \operatorname{grad}(T)$ becomes the bilinear form in the finite element formulation.

   Weak Form Foundation

   When integrated over a domain Ω, this relationship becomes:

$$\int_\Omega V \ \mathrm{div}(\mathbf{K}\ \mathrm{grad}(T))\, d\Omega = \int_\Omega \mathrm{div}(V\,\mathbf{K}\ \mathrm{grad}(T))\, d\Omega - \int_\Omega \mathrm{grad}(V) \cdot \mathbf{K} \cdot \mathrm{grad}(T)\, d\Omega$$

Using the divergence theorem, the first integral on the right becomes a boundary integral, leading to the standard weak form used in finite element analysis.

FEM in 2/3D

2D/3D Linear Heat Equation: Strong Form to Weak Form

Problem Setup

Governing equation: $\mathrm{div}(\mathbf{q}) = 0$ where we assumed that there is no source term for simplicity.

Constitutive equation (Fourier's law): $\mathbf{q} = -\mathbf{K} \cdot \mathrm{grad}(T)$

Boundary conditions: - $\Gamma_{T_0} : T = T_0$ (Essential BC) - $\Gamma_{T_1} : T = T_1$ (Essential BC)
- $\Gamma_q : \mathbf{q} = \mathbf{q}_n^*$ (Natural BC)
where $\mathbf{q}_n^*$ is the prescribed heat flux on the boundary Γ_q in the direction of the (outwards) normal to the surface.

Combined form: $\mathrm{div}(\mathbf{q}) = -\mathrm{div}(\mathbf{K} \cdot \mathrm{grad}(T)) = 0$

Weak Form Derivation

Step 1: Multiply by test function V and integrate over domain

$$\int_{\Omega} \mathrm{div}(\mathbf{K} \cdot \mathrm{grad}(T))V \, d\Omega = 0$$

Step 2: Apply the product rule result

$$\int_{\Omega} \mathrm{div}(V\mathbf{K} \cdot \mathrm{grad}(T)) \, d\Omega - \int_{\Omega} \mathrm{grad}(V) \cdot \mathbf{K} \cdot \mathrm{grad}(T) \, d\Omega = 0$$

Step 3: Apply divergence theorem to first integral

$$\int_{\Omega} \mathrm{div}(V\mathbf{K} \cdot \mathrm{grad}(T)) \, d\Omega = \int_{\Omega} \mathrm{div}(V\mathbf{q}) \, d\Omega = \int_{\partial\Omega} V\mathbf{q} \cdot \mathbf{n} \, d\Gamma$$

Final weak form:

$$\int_{\partial\Omega} V\mathbf{q} \cdot \mathbf{n} \, d\Gamma - \int_{\Omega} \mathrm{grad}(V) \cdot \mathbf{K} \cdot \mathrm{grad}(T) \, d\Omega = 0$$

Finite element approximation:

$$V_h = \sum_{i=1}^{N} b_i \phi_i(x, y), \quad T_h = \sum_{j=1}^{N} a_j \phi_j(x, y) \Rightarrow \{a_j\}$$

2D/3D Finite Elements: Geometry and Shape Functions

Transition from 1D to 2D/3D

Key differences:

- **1D:** Geometric description is simpler - element domain is simply a line
- **2D/3D:** Geometry plays a crucial role and complexity increases significantly

Most common finite element geometries:

- **2D:** Triangular elements & Rectangular elements
- **3D:** Tetrahedron elements & Hexahedron elements

2D Rectangular Elements: Linear Shape Functions

Design requirements for shape functions:

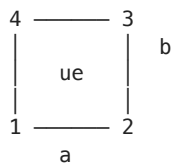
- **Simplicity:** Describe change in the element's domain with minimal complexity
- **Local support:** Each shape function corresponds to one node, value of "1" at one node, "0" at all other nodes

1D analogy: $u^e = a_i \phi_i^g + a_{i+1} \phi_{i+1}^g$ (where $u^e \in [x_i, x_{i+1}]$)

2D extension: For a linear rectangular element, we need **four shape functions** to describe u^e

Rectangular Element with Linear Shape Functions

Consider a rectangular element with nodes numbered as follows:



Shape functions for specific rectangular element:

$\phi_1^g = \frac{1}{ab}(a-x)(b-y)$

$\phi_2^g = \frac{1}{ab}x(b-y)$

$\phi_3^g = \frac{1}{ab}xy$

$\phi_4^g = \frac{1}{ab}(a-x)y$

Verification : - $\phi_1^g(0,0) = 1, \phi_1^g(a,0) = 0, \phi_1^g(0,b) = 0, \phi_1^g(a,b) = 0$ ✓

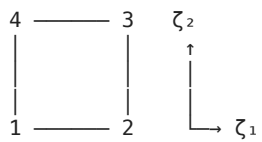
- $\phi_2^g(0,0) = 0, \phi_2^g(a,0) = 1, \phi_2^g(0,b) = 0, \phi_2^g(a,b) = 0$ ✓

Element solution: $u^e = a_1\phi_1^g + a_2\phi_2^g + a_3\phi_3^g + a_4\phi_4^g = u^e(x,y)$

Master Element Concept in 2D

Mapping to Master Element

Purpose: Map general elements to a standardized **master element** for easier computation
Coordinate transformation: $[x, y] \longrightarrow [\zeta_1, \zeta_2] \in [-1, 1] \times [-1, 1]$
Master element node numbering convention:



Master element shape functions:

$$\hat{\phi_1} = \phi_1^e(\zeta_1, \zeta_2) = \frac{1}{4}(1 - \zeta_1)(1 - \zeta_2)$$
$$\hat{\phi_2} = \phi_2^e(\zeta_1, \zeta_2) = \frac{1}{4}(1 + \zeta_1)(1 - \zeta_2)$$
$$\hat{\phi_3} = \phi_3^e(\zeta_1, \zeta_2) = \frac{1}{4}(1 + \zeta_1)(1 + \zeta_2)$$
$$\hat{\phi_4} = \phi_4^e(\zeta_1, \zeta_2) = \frac{1}{4}(1 - \zeta_1)(1 + \zeta_2)$$

Bi-linear nature: Each $\hat{\phi_i}$ equals “1” at the i-th node and “0” at other nodes, creating **bi-linear** shape functions.

Example expansion: $\hat{\phi}_1 = \frac{1}{4}(1-\zeta_1)(1-\zeta_2) = \frac{1}{4}(1-\zeta_1-\zeta_2+\zeta_1\zeta_2)$
The term $\zeta_1\zeta_2$ makes it bi-linear (linear in both coordinates).

Jacobian and Coordinate Transformation

Jacobian Matrix

Definition:

$$\mathbf{F} = \begin{bmatrix} \frac{\partial x}{\partial \zeta_1} & \frac{\partial x}{\partial \zeta_2} \\ \frac{\partial y}{\partial \zeta_1} & \frac{\partial y}{\partial \zeta_2} \end{bmatrix}, \quad J = \mathrm{Det}(\mathbf{F}) = \frac{\partial x}{\partial \zeta_1}\frac{\partial y}{\partial \zeta_2} - \frac{\partial x}{\partial \zeta_2}\frac{\partial y}{\partial \zeta_1}$$

Critical requirement: $J > 0$ throughout the element for a “good” mapping

Global to Local Mapping

we will use **isoparametric** mapping, where the same shape functions are used for both geometry and field variables.

Coordinate transformation:

$$x(\chi_1, \chi_2) = \sum_{i=1}^{4} \chi_{1i}\hat{\phi_i} = \chi_{11}\hat{\phi_1} + \chi_{12}\hat{\phi_2} + \chi_{13}\hat{\phi_3} + \chi_{14}\hat{\phi_4}$$

$$y(\chi_1, \chi_2) = \sum_{i=1}^{4} \chi_{2i}\hat{\phi_i}$$

Where χ_{1i} are global x-coordinates and χ_{2i} are global y-coordinates of element nodes.

Element Quality: Good vs Bad Elements

Acceptable Elements

Case 1 - Rectangular element:
$0 < J(\zeta_1, \zeta_2) < \infty$ throughout element - Jacobian is constant - ✅ **Acceptable**

Case 3 - Distorted but convex:
$0 < J(\zeta_1, \zeta_2) < \infty$ throughout element - Jacobian varies but remains positive and bounded - ✅ **Acceptable**

Unacceptable Elements

Case 2 - Incorrect node numbering:
$J(\zeta_1, \zeta_2) < 0$ throughout element - Nodes numbered incorrectly, turning element "inside out" - ❌ **Unacceptable**

Case 4 - Partially negative Jacobian:
$J(\zeta_1, \zeta_2) < 0$ in some regions - Can cause

singularities in stiffness matrix - ❌
Unacceptable

Key insight for linear elements: The primary indicator of problematic elements is **non-convexity**, even with correct numbering.

Bookkeeping in 2D Finite Elements

Difference from 1D

1D: Node numbering is intuitive and element connectivity is straightforward

1 — 2 — 3 — 4 — 5 — 6 — 7 — 8 — 9

2D: More complex connectivity patterns require systematic bookkeeping

Element Connectivity Tables

Element-to-node connectivity:

Element	Node 1	Node 2	Node 3	Node 4
1	20	21	2	1
2	21	22	3	2

Node coordinate table:

Node	X-coord	Y-coord
1	X_1	Y_1
2	X_2	Y_2
…	…	…

Importance: Essential for assembling the global stiffness matrix $[K_{ij}^{g}]$ and load vector $\{F_i^g\}$, and for enforcing boundary conditions.

Shape Functions: Differential Properties

Coordinate Transformation Relations

Forward transformation (ζ → x):

$$\frac{\partial}{\partial \zeta_1} = \frac{\partial}{\partial x}\frac{\partial x}{\partial \zeta_1} + \frac{\partial}{\partial y}\frac{\partial y}{\partial \zeta_1}$$
$$\frac{\partial}{\partial \zeta_2} = \frac{\partial}{\partial x}\frac{\partial x}{\partial \zeta_2} + \frac{\partial}{\partial y}\frac{\partial y}{\partial \zeta_2}$$

Inverse transformation (x → ζ):

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \zeta_1}\frac{\partial \zeta_1}{\partial x} + \frac{\partial}{\partial \zeta_2}\frac{\partial \zeta_2}{\partial x}$$
$$\frac{\partial}{\partial y} = \frac{\partial}{\partial \zeta_1}\frac{\partial \zeta_1}{\partial y} + \frac{\partial}{\partial \zeta_2}\frac{\partial \zeta_2}{\partial y}$$

Matrix form:

$$\begin{bmatrix} dx \\ dy \end{bmatrix} = \mathbf{F}\begin{bmatrix} d\zeta_1 \\ d\zeta_2 \end{bmatrix}, \qquad \begin{bmatrix} d\zeta_1 \\ d\zeta_2 \end{bmatrix} = \mathbf{F}^{-1}\begin{bmatrix} dx \\ dy \end{bmatrix}$$

Gradient Calculation in Master Element

For test function V:

$$\mathrm{grad}(V) = \begin{bmatrix} \frac{\partial V}{\partial x} \\ \frac{\partial V}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{\partial \zeta_1}{\partial x} & \frac{\partial \zeta_2}{\partial x} \\ \frac{\partial \zeta_1}{\partial y} & \frac{\partial \zeta_2}{\partial y} \end{bmatrix}^T \begin{bmatrix} \frac{\partial V}{\partial \zeta_1} \\ \frac{\partial V}{\partial \zeta_2} \end{bmatrix}$$

For temperature field T:

$$\mathrm{grad}(T) = \begin{bmatrix} \frac{\partial T}{\partial x} \\ \frac{\partial T}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{\partial \zeta_1}{\partial x} & \frac{\partial \zeta_2}{\partial x} \\ \frac{\partial \zeta_1}{\partial y} & \frac{\partial \zeta_2}{\partial y} \end{bmatrix}^T \begin{bmatrix} \frac{\partial T}{\partial \zeta_1} \\ \frac{\partial T}{\partial \zeta_2} \end{bmatrix}$$

Extension to 3D Vector Problems

Linear Elasticity Example

Governing equation: $\mathrm{div}(\boldsymbol{\sigma}) + \mathbf{f} = 0$
Constitutive relation: $\boldsymbol{\sigma} = \mathbf{E} \cdot \boldsymbol{\varepsilon} = \mathbf{E} \cdot \mathrm{grad\hat{}s}(\mathbf{u})$
Divergence theorem in 3D:

$$\int_{\Omega} \mathrm{div}(\boldsymbol{\sigma})\, d\Omega = \int_{\partial\Omega} \boldsymbol{\sigma} \cdot \mathbf{n}\, da$$

Key extensions for 3D:

- **Vector unknowns:** Displacement field $\mathbf{u} = [u_x, u_y, u_z]^T$
- **Matrix operations:** Stress and strain tensors
- **Global/local transformations:** More complex connectivity
- **Vector problem connectivity:** Multiple DOFs per node

3D element example

The B Matrix: Relating Nodal Values to Field Gradients

In the Finite Element Method, we approximate a continuous field (like temperature T or displacement u) within an element using shape functions N_j and nodal values. For example, the temperature T(x,y) at any point within an element can be approximated as:

$$T(x,y) \approx \sum_{j=1}^{n_{\text{nodes}}} N_j(x,y)a_j = \mathbf{N}\mathbf{a}^e$$

where $N_j(x,y)$ are the shape functions, a_j are the nodal temperatures for the element, $\mathbf{N}$ is the row vector of shape functions, and $\mathbf{a}^e$ is the column vector of nodal temperatures for the element.

The physical behavior of the system (e.g., heat flux, stress/strain) often depends on the gradient of this field. The B matrix is a cornerstone of FEM as it directly relates the nodal values of an element to the gradient of the field (or its derivatives) within that element.

For Scalar Field Problems (e.g., Heat Conduction)

In heat conduction, the heat flux $\mathbf{q}$ is related to the temperature gradient $\mathrm{grad}(T)$ by Fourier's Law: $\mathbf{q} = -\mathbf{K} \cdot \mathrm{grad}(T)$.
The weak form integral that leads to the stiffness matrix, $\int_{\Omega^e} \mathrm{grad}(V) \cdot \mathbf{K} \cdot \mathrm{grad}(T)\, d\Omega$, requires evaluating these gradients.
Let $T = \mathbf{N}\mathbf{a}^e$. The temperature gradient in 2D is:

$$\mathrm{grad}(T) = \begin{Bmatrix} \frac{\partial T}{\partial x} \\ \frac{\partial T}{\partial y} \end{Bmatrix} = \begin{Bmatrix} \frac{\partial(\sum N_j a_j)}{\partial x} \\ \frac{\partial(\sum N_j a_j)}{\partial y} \end{Bmatrix} = \sum_{j=1}^{n_{\text{nodes}}} \begin{bmatrix} \frac{\partial N_j}{\partial x} \\ \frac{\partial N_j}{\partial y} \end{bmatrix} a_j$$

This can be written in matrix form as:

$$\mathrm{grad}(T) = \mathbf{B}\mathbf{a}^e$$

Where the B matrix for a 2D scalar problem (like heat conduction) for an element with n_{nodes} is:

$$\mathbf{B} = \begin{bmatrix} \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial x} & \cdots & \frac{\partial N_{n_{\text{nodes}}}}{\partial x} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_2}{\partial y} & \cdots & \frac{\partial N_{n_{\text{nodes}}}}{\partial y} \end{bmatrix}$$

Each column j of the B matrix corresponds to node j and contains the partial derivatives of its shape function N_j with respect to the global coordinates. For a 3D scalar problem, another row for $\frac{\partial N_j}{\partial z}$ would be added.

Role in Element Stiffness Matrix:

The element stiffness matrix for heat conduction, derived from the weak form, is given by:

$$[\mathbf{k}^e] = \int_{\Omega^e} \mathbf{B}^T \mathbf{K} \mathbf{B} \, d\Omega$$

Here, $\mathbf{K}$ is the material property matrix (thermal conductivity tensor). For an isotropic material in 2D with scalar conductivity k, $\mathbf{K} = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$. The integration is performed over the volume (or area in 2D) of the element Ω^e.

Calculation of B Matrix using Master Element Coordinates:

Shape functions N_j are typically defined in local (master element) coordinates (ζ_1, ζ_2) (often denoted ξ, η). Their derivatives with respect to global coordinates (x,y) are found using the chain rule and the Jacobian matrix $\mathbf{F}$ of the coordinate transformation:

$$\begin{Bmatrix} \frac{\partial N_j}{\partial x} \\ \frac{\partial N_j}{\partial y} \end{Bmatrix} = \mathbf{F}^{-T} \begin{Bmatrix} \frac{\partial N_j}{\partial \zeta_1} \\ \frac{\partial N_j}{\partial \zeta_2} \end{Bmatrix}$$

where $\mathbf{F}^{-T}$ is the inverse transpose of the Jacobian matrix $\mathbf{F}$, defined as:

where $\mathbf{F}^{-T} = (\mathbf{F}^{-1})^T$.

The Jacobian matrix $\mathbf{F}$ is defined as $\mathbf{F} = \begin{bmatrix} \frac{\partial x}{\partial \zeta_1} & \frac{\partial x}{\partial \zeta_2} \\ \frac{\partial y}{\partial \zeta_1} & \frac{\partial y}{\partial \zeta_2} \end{bmatrix}$.

The derivatives $\frac{\partial N_j}{\partial \zeta_1}$ and $\frac{\partial N_j}{\partial \zeta_2}$ are easily computed from the master element shape function definitions. The components of $\mathbf{F}^{-T}$ depend on the specific element geometry.

For Vector Field Problems (e.g., Elasticity)

The concept of the B matrix is fundamental in structural mechanics (elasticity), where it relates nodal displacements to strains within an element.

The displacement field $\mathbf{u}$ is approximated as $\mathbf{u} = \mathbf{N}\mathbf{d}^e$, where $\mathbf{d}^e$ is the vector of all nodal displacements for the element, and $\mathbf{N}$ is the matrix of shape functions arranged appropriately for a vector field.

The strain $\boldsymbol{\varepsilon}$ is obtained by differentiating the displacement field: $\boldsymbol{\varepsilon} = \mathbf{L}\mathbf{u}$, where $\mathbf{L}$ is a differential operator matrix. Combining these gives:

$$\boldsymbol{\varepsilon} = \mathbf{L}(\mathbf{N}\mathbf{d}^e) = (\mathbf{L}\mathbf{N})\mathbf{d}^e = \mathbf{B}\mathbf{d}^e$$

The B matrix in elasticity thus contains derivatives of shape functions. The element stiffness matrix is then computed as:

$$[\mathbf{k}^e] = \int_{\Omega^e} \mathbf{B}^T \mathbf{D} \mathbf{B} \, d\Omega$$

where $\mathbf{D}$ is the material elasticity matrix (constitutive matrix relating stress to strain).

B Matrix for 2D Elasticity (Plane Stress / Plane Strain)

For a 2D elasticity problem, the displacement at any point within an element is

$$\mathbf{u}(x,y) = \begin{Bmatrix} u(x,y) \\ v(x,y) \end{Bmatrix}.$$

The nodal displacement vector for a node i is $\mathbf{d}_i = \begin{Bmatrix} u_i \\ v_i \end{Bmatrix}$.

The displacement field is interpolated as:

$$\begin{Bmatrix} u \\ v \end{Bmatrix} = \sum_{i=1}^{n_{\text{nodes}}} \begin{bmatrix} N_i(x,y) & 0 \\ 0 & N_i(x,y) \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \end{Bmatrix}$$

$$\boldsymbol{\varepsilon} = \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{Bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \end{Bmatrix}$$

$$\mathbf{B}_i = \begin{bmatrix} \frac{\partial N_i}{\partial x} & 0 \\ 0 & \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} \end{bmatrix}$$

$$\mathbf{B} = [\mathbf{B}_1 \quad \mathbf{B}_2 \quad \dots \quad \mathbf{B}_{n_{\text{nodes}}}]$$

So, if an element has n_{nodes} nodes, the B matrix will have 3 rows (for $\varepsilon_x, \varepsilon_y, \gamma_{xy}$) and $2 \times n_{\text{nodes}}$ columns.

B Matrix for 3D Elasticity

For a 3D elasticity problem, the displacement at any point is $\mathbf{u}(x, y, z) = \begin{Bmatrix} u(x, y, z) \\ v(x, y, z) \\ w(x, y, z) \end{Bmatrix}$.

The nodal displacement vector for node i is $\mathbf{d}_i = \begin{Bmatrix} u_i \\ v_i \\ w_i \end{Bmatrix}$.

The displacement field is interpolated as:

$$\begin{Bmatrix} u \\ v \\ w \end{Bmatrix} = \sum_{i=1}^{n_{\text{nodes}}} \begin{bmatrix} N_i(x, y, z) & 0 & 0 \\ 0 & N_i(x, y, z) & 0 \\ 0 & 0 & N_i(x, y, z) \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ w_i \end{Bmatrix}$$

The strain vector (engineering strains) in 3D is:

$$\boldsymbol{\varepsilon} = \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = \begin{Bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} \\ \frac{\partial w}{\partial z} \\ \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \\ \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \\ \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \end{Bmatrix}$$

For each node, its contribution to the strain is through. The matrix relates the strains to the nodal displacements of node:

$$\mathbf{B}_i = \begin{bmatrix} \frac{\partial N_i}{\partial x} & 0 & 0 \\ 0 & \frac{\partial N_i}{\partial y} & 0 \\ 0 & 0 & \frac{\partial N_i}{\partial z} \\ \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} & 0 \\ 0 & \frac{\partial N_i}{\partial z} & \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial z} & 0 & \frac{\partial N_i}{\partial x} \end{bmatrix}$$

The full element B matrix is assembled as $\mathbf{B} = [\mathbf{B}_1 \quad \mathbf{B}_2 \quad \ldots \quad \mathbf{B}_{n_{\text{nodes}}}]$.

If an element has n_{nodes} nodes, the B matrix will have 6 rows and $3 \times n_{\text{nodes}}$ columns. The derivatives $\frac{\partial N_i}{\partial x}$, $\frac{\partial N_i}{\partial y}$, and $\frac{\partial N_i}{\partial z}$ are again found using the Jacobian transformation from the master element coordinates.

Key Takeaway (B Matrix)

The B matrix is a crucial component in finite element analysis.
It contains the spatial derivatives of the element shape functions, effectively translating the discrete nodal degrees of freedom into continuous gradient fields (like temperature gradients) or deformation measures (like mechanical strains) within an element.
This matrix is essential for forming the element stiffness matrix and, consequently, for solving the overall system of equations that describes the physical problem.
The components of B are generally functions of position within the element (unless using simple elements like constant strain triangles), and are evaluated numerically during the integration process to form the element stiffness matrix.

Shape Functions: Differential Properties

To evaluate terms like $\mathrm{grad}(V)$ and $\mathrm{grad}(T)$ in the weak form, we need the derivatives of shape functions (and consequently, field variables) with respect to global coordinates (x,y). However, shape functions $N(\zeta_1, \zeta_2)$ are defined in the master element coordinates (ζ_1, ζ_2).

We use the chain rule and the Jacobian of the coordinate mapping to transform these derivatives.

Coordinate Transformation and the Jacobian Matrix

Recall the isoparametric mapping from master coordinates (ζ_1, ζ_2) to global coordinates (x,y):

$$x = \sum_{i=1}^{n_{\text{nodes}}} N_i(\zeta_1, \zeta_2) x_i$$

$$y = \sum_{i=1}^{n_{\text{nodes}}} N_i(\zeta_1, \zeta_2) y_i$$

The Jacobian matrix of this transformation is:

$$\mathbf{F} = \begin{bmatrix} \frac{\partial x}{\partial \zeta_1} & \frac{\partial x}{\partial \zeta_2} \\ \frac{\partial y}{\partial \zeta_1} & \frac{\partial y}{\partial \zeta_2} \end{bmatrix}$$

The determinant of this matrix, $J = \det(\mathbf{F})$, is used in changing variables for integration: $dx\,dy = J\,d\zeta_1\,d\zeta_2$.

Differentials are related by:

$$\begin{Bmatrix} dx \\ dy \end{Bmatrix} = \mathbf{F} \begin{Bmatrix} d\zeta_1 \\ d\zeta_2 \end{Bmatrix}$$

$$\begin{Bmatrix} d\zeta_1 \\ d\zeta_2 \end{Bmatrix} = \mathbf{F}^{-1} \begin{Bmatrix} dx \\ dy \end{Bmatrix}$$

$$\mathbf{F}^{-1} = \begin{bmatrix} \frac{\partial \zeta_1}{\partial x} & \frac{\partial \zeta_1}{\partial y} \\ \frac{\partial \zeta_2}{\partial x} & \frac{\partial \zeta_2}{\partial y} \end{bmatrix}$$

Transformation of Derivatives

We need to express derivatives with respect to global coordinates (x,y) in terms of derivatives with respect to master coordinates (ζ_1, ζ_2), since our shape functions N_i are given as $N_i(\zeta_1, \zeta_2)$.

Consider a function $V(\zeta_1, \zeta_2)$. Its derivatives with respect to ζ_1 and ζ_2 can be related to its derivatives with respect to x and y using the chain rule:

$$\frac{\partial V}{\partial \zeta_1} = \frac{\partial V}{\partial x}\frac{\partial x}{\partial \zeta_1} + \frac{\partial V}{\partial y}\frac{\partial y}{\partial \zeta_1}$$

$$\frac{\partial V}{\partial \zeta_2} = \frac{\partial V}{\partial x}\frac{\partial x}{\partial \zeta_2} + \frac{\partial V}{\partial y}\frac{\partial y}{\partial \zeta_2}$$

$$\left\{ \begin{array}{c} \frac{\partial V}{\partial \zeta_1} \\ \frac{\partial V}{\partial \zeta_2} \end{array} \right\} = \left[\begin{array}{cc} \frac{\partial x}{\partial \zeta_1} & \frac{\partial y}{\partial \zeta_1} \\ \frac{\partial x}{\partial \zeta_2} & \frac{\partial y}{\partial \zeta_2} \end{array} \right] \left\{ \begin{array}{c} \frac{\partial V}{\partial x} \\ \frac{\partial V}{\partial y} \end{array} \right\} = \mathbf{F}^T \left\{ \begin{array}{c} \frac{\partial V}{\partial x} \\ \frac{\partial V}{\partial y} \end{array} \right\}$$

$$\left\{ \begin{array}{c} \frac{\partial V}{\partial x} \\ \frac{\partial V}{\partial y} \end{array} \right\} = (\mathbf{F}^T)^{-1} \left\{ \begin{array}{c} \frac{\partial V}{\partial \zeta_1} \\ \frac{\partial V}{\partial \zeta_2} \end{array} \right\} = \mathbf{F}^{-T} \left\{ \begin{array}{c} \frac{\partial V}{\partial \zeta_1} \\ \frac{\partial V}{\partial \zeta_2} \end{array} \right\}$$

$$\mathbf{F}^{-T} = \left(\left[\begin{array}{cc} \frac{\partial x}{\partial \zeta_1} & \frac{\partial x}{\partial \zeta_2} \\ \frac{\partial y}{\partial \zeta_1} & \frac{\partial y}{\partial \zeta_2} \end{array} \right] \right)^T = \left[\begin{array}{cc} \frac{\partial x}{\partial \zeta_1} & \frac{\partial y}{\partial \zeta_1} \\ \frac{\partial x}{\partial \zeta_2} & \frac{\partial y}{\partial \zeta_2} \end{array} \right]$$

Gradient Calculation in Master Element Coordinates

Using the relationship derived above, the gradient of a test function V (or temperature field T) in global coordinates is:

$$\text{grad}(V) = \left\{ \begin{array}{c} \frac{\partial V}{\partial x} \\ \frac{\partial V}{\partial y} \end{array} \right\} = \mathbf{F}^{-T} \left\{ \begin{array}{c} \frac{\partial V}{\partial \zeta_1} \\ \frac{\partial V}{\partial \zeta_2} \end{array} \right\} = \left[\begin{array}{cc} \frac{\partial x}{\partial \zeta_1} & \frac{\partial y}{\partial \zeta_1} \\ \frac{\partial x}{\partial \zeta_2} & \frac{\partial y}{\partial \zeta_2} \end{array} \right] \left\{ \begin{array}{c} \frac{\partial V}{\partial \zeta_1} \\ \frac{\partial V}{\partial \zeta_2} \end{array} \right\}$$

$$\frac{\partial V}{\partial x} = \frac{\partial x}{\partial \zeta_1} \frac{\partial V}{\partial \zeta_1} + \frac{\partial x}{\partial \zeta_2} \frac{\partial V}{\partial \zeta_2}$$

$$\frac{\partial V}{\partial y} = \frac{\partial y}{\partial \zeta_1} \frac{\partial V}{\partial \zeta_1} + \frac{\partial y}{\partial \zeta_2} \frac{\partial V}{\partial \zeta_2}$$

To implement this:

1. For each shape function $N_i(\zeta_1, \zeta_2)$, calculate its derivatives $\frac{\partial N_i}{\partial \zeta_1}$ and $\frac{\partial N_i}{\partial \zeta_2}$ analytically in the master element.

2. At each integration point (ζ_1, ζ_2) within the master element:

a. Calculate the Jacobian matrix $\mathbf{F}$ using the derivatives of the mapping functions $x(\zeta_1, \zeta_2)$ and $y(\zeta_1, \zeta_2)$.

b. Compute $\mathbf{F}^{-1}$ and then $\mathbf{F}^{-T}$.

c. Use $\mathbf{F}^{-T}$ to transform the local derivatives $\left\{ \begin{array}{c} \frac{\partial N_i}{\partial \zeta_1} \\ \frac{\partial N_i}{\partial \zeta_2} \end{array} \right\}$ into global derivatives $\left\{ \begin{array}{c} \frac{\partial N_i}{\partial x} \\ \frac{\partial N_i}{\partial y} \end{array} \right\}$.

These global derivatives form the columns of the B matrix for scalar problems, or parts of the columns for vector problems.

Summary: Key Concepts

- **Mathematical Foundation:**
- Product rule for divergence enables weak form derivation
- Divergence theorem converts domain integrals to boundary integrals
- Integration by parts transfers derivatives to test functions
- **Geometric Considerations:**
- Master element concept simplifies integration
- Jacobian must be positive for valid elements
- Element quality depends on convexity and node ordering
- **Implementation Details:**
- Shape functions provide local approximation
- Connectivity tables manage global assembly
- Coordinate transformations enable numerical integration
- **Extension to Higher Dimensions:**

- Same mathematical principles apply
- Increased complexity in bookkeeping and assembly
- Vector problems require multiple DOFs per node

Speaker notes